SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA0510**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks50** **Duration :60 Minutes**

Annexure A

Analytical Problem Solving

Code of Conduct:

I Premarubha N(192525020) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Premarubha N

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.
(b) Apply a negative transformation $s=255-r$ to the pixel values:
$r=[50,100,150,200]$
Compute the transformed values.

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.
(b) Given $s=c\log(1+r)$, where $c=10$, compute s for:
$r=[0,10,50,100]$

3. Histogram Processing

(a) Explain the role of histogram equalization in contrast enhancement.
(b) Given a 3-bit image with histogram:

Intensity	**Frequency**
0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.
(b) Apply a 3×3 averaging filter to the following region:
10 20 30 20 30 40 30 40 50
Compute the new center pixel value.

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.
(b) Apply the Laplacian mask:
[0 -1 0}, {-1 4 -1], { 0 -1 0]
to the matrix:
10 10 10, 10 20 10, 10 10 10
Compute the output at the center.

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ITA0510 – COMPUTER VISION
CO2 ASSESSMENT TOOL 1

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1. Grey Level Transformation

(a) Purpose of grey level transformation

Grey level transformation is used in image enhancement to modify the intensity values of pixels. It helps to improve image contrast, highlight important details, make dark or bright regions easier to observe, and convert an image into its negative when required.

(b) Negative transformation

Given: $s = 255 - r$

Pixel values: $r = [50, 100, 150, 200]$

For $r = 50$: $s = 255 - 50 = 205$

For $r = 100$: $s = 255 - 100 = 155$

For $r = 150$: $s = 255 - 150 = 105$

For $r = 200$: $s = 255 - 200 = 55$

Answer: $s = [205, 155, 105, 55]$

2. Log Transformation

a. Log transformation is given by $s = c \log(1 + r)$. It expands the values of low-intensity pixels while compressing high-intensity values. Therefore, details present in darker portions of an image become more visible. It is useful for images having a large range of intensity values.

Given: $s = 10 \log(1 + r)$, using natural logarithm (ln)

For $r = 0$: $s = 10 \ln(1) = 0$

For $r = 10$: $s = 10 \ln(11) = 10(2.3979) = 23.979$

For $r = 50$: $s = 10 \ln(51) = 10(3.9318) = 39.318$

For $r = 100$: $s = 10 \ln(101) = 10(4.6151) = 46.151$

Answer: $s = [0, 23.98, 39.32, 46.15]$

3. Histogram Processing

(a) Role of histogram equalization

Histogram equalization is a technique used to improve the contrast of an image. It redistributes pixel intensity values over a wider range. As a result, dark regions become clearer, bright regions become more distinguishable, hidden details can become visible, and overall contrast is improved.

(b) Given histogram

Intensity	Frequency	Probability	CDF
0	2	0.125	0.125
1	2	0.125	0.250
2	4	0.250	0.500
3	8	0.500	1.000

Total pixels: $N = 2 + 2 + 4 + 8 = 16$

For a 3-bit image: $L = 2^3 = 8$, so intensity levels range from 0 to 7.

Equalized intensity: $s = (L - 1) \times CDF = 7 \times CDF$

$s_0 = 7(0.125) = 0.875 \approx 1$ $s_1 =$
$7(0.25) = 1.75 \approx 2$ $s_2 = 7(0.50) =$
$3.5 \approx 4$ $s_3 = 7(1) = 7$

Answer: Equalized intensity values = [1, 2, 4, 7]

4. **Spatial Filtering (Smoothing)**

(a) Purpose

Spatial filtering for smoothing is used to reduce noise, remove small unwanted variations, blur sharp intensity changes, and produce a smoother image. A 3 × 3 averaging filter replaces the center pixel with the average of the nine pixels in its neighborhood.

(b) Given region

10	20	30
20	30	40
30	40	50

Using a 3 × 3 averaging filter:

Sum = 10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270

New center pixel = 270 / 9 = 30

Answer: New center pixel = 30

5. Spatial Filtering (Sharpening)

(a) How sharpening enhances image details

Sharpening enhances edges and fine details in an image by emphasizing rapid changes in pixel intensity. It is useful for edge enhancement, making objects more distinct, highlighting fine details, and improving boundaries between different regions.

(b) Laplacian mask

Mask:

0 −1 0

−1 4 −1

0 −1 0

Image region:

10 10 10

10 20 10

10 10 10

Applying the mask at the center:

(0)(10) + (−1)(10) + (0)(10) + (−1)(10) + (4)(20) + (−1)(10) + (0)(10) + (−1)(10) +

(0)(10)

= −10 − 10 + 80 − 10 − 10

= 40

Answer: Laplacian output at center = 40

If the final sharpened center pixel is required using $g = f + \nabla^2 f$:

$20 + 40 = 60$

Therefore, Laplacian response = 40 and sharpened center value = 60.